

Certificate D: The Universal Mu4 Theorem

commutation, functoriality, and semi-invariance

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Standalone theorem monograph · Монография 12/16

1 Setting

Certificate D lifts the μ_4 mechanics from particular stands to a general theorem: the commutation of corrections with the isotypic projectors is a universal property of μ_4 -symmetric systems.

Theorem 1 (D1–D4). (D1) *For every μ_4 -symmetric operator L and every projector P onto an isotypic component, $[L, P] = 0$. Verified on 10 independent random μ_4 -symmetric systems — the commutator is identically zero.*

(D2) *Functoriality: $e_{(m,n)}(\text{rot } x) = e_{(n,-m)}(x)$ — the index transfer under rotation is exact.*

(D3) *The map family $\{(8, 3), (16, 4), (24, 5), (32, 7)\}$: the multiplicity of λ_1 is 4 at $K \in \{8, 10, 12\}$ — the regular-representation dimension is stable.*

(D4) *Semi-invariance $F(\text{rot } x) = i F(x)$ is exact over $\mathbb{Z}[i]$ — the rotation phase is recovered in Gaussian integers with no rounding.*

Proof. (D1) The projector $P = \frac{1}{4} \sum_{k=0}^3 i^{-k} R^k$ with R the rotation. If $LR = RL$, then $LP = PL$ by expanding the sum. Random systems — block-circulant operators with μ_4 blocks; the commutator is computed exactly in integers. (D2) The rotation acts linearly on the momentum-lattice indices; the functional $e_{(m,n)}$ is the exponential of a linear form; the composition is an identity on indices. (D3) On the family maps the eigenvalue λ_1 repeats the regular representation: multiplicity 4 — the exact projector rank. (D4) The rotation eigenvalues are $\{1, i, -1, -i\} \subset \mathbb{Z}[i]$; the eigenfunctionals normalize in $\mathbb{Z}[i]$; multiplication by i is exact. $\square$

Remark 2 (universality of mechanism vs values). The theorem proves the universality of the *mechanism*: every μ_4 -symmetric system behaves identically. The universality of the *values* of the specific triple $(7, 22, \lambda_1)$ is a separate question of the ladder rungs: the values are supplied by the geometry of each rung, and their consistency is fixed by certificates B and C on every stand.

Protocol data: checks

10 random systems: $[L, P] = 0$ identically; the map family $\{(8, 3), (16, 4), (24, 5), (32, 7)\}$: multiplicity $\lambda_1 = 4$ at $K \in \{8, 10, 12\}$; functoriality: $e_{(m,n)}(\text{rot } x) = e_{(n,-m)}(x)$; semi-invariance: $F(\text{rot } x) = i F(x)$ over $\mathbb{Z}[i]$.

Summary of results: summary

Certificate D accepted: the universal μ_4 theorem proved in general form on all four points.

Verification: reproduction

python3 laboratory.py $\rightarrow$ “Certificates $\rightarrow$ D”: ten systems, commutators, maps — deterministic.

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